MTH101, Basic Linear Algebra

Monsoon 2025

18/09/2025, 2:00PM-3:00PM.

Tutorial-3

(1) Let $A = \begin{pmatrix} 1 & 2 & 1 \\ 2 & 3 & 1 \\ 1 & 4 & 1 \end{pmatrix}$ and $B = \begin{pmatrix} 1 & -4 & 1 \\ -3 & 2 & -1 \\ 7 & -4 & 1 \end{pmatrix}$. Calculate $A + B$ and AB.

(2) Let P, Q, R be $n \times n$ matrices such that $PQ = I_n$ and $RP = I_n$. Show that $R = Q$.

(3) Solve the system of equations using row reduction

(i)

$$\begin{aligned} 2X + 7Y &= 43 \\ 5X + 3Y &= 35 \end{aligned}$$

(ii)

$$\begin{aligned} -3X + 7Y + 2Z &= 4 \\ 4X - 2Y + Z &= 7 \\ -2X + 12Y + 5Z &= 15 \end{aligned}$$

(iii)

$$\begin{aligned} X_1 + 3X_2 + 7X_3 + 2X_4 &= 14 \\ 4X_2 + X_3 - X_4 &= -3 \\ 2X_1 + 3X_3 + X_4 &= 5 \\ X_1 + X_2 + 3X_3 &= 2 \end{aligned}$$